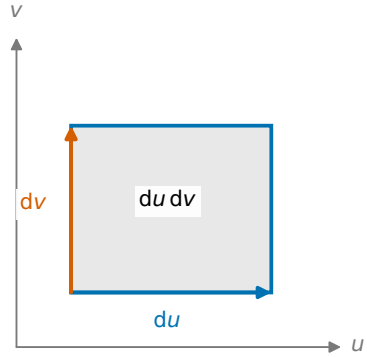
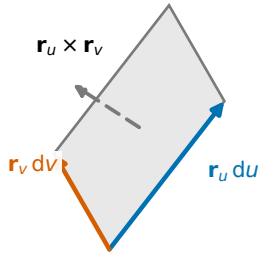
A



$(u, v) \mapsto \mathbf{r}(u, v)$

B



$dS = \|\mathbf{r}_u \times \mathbf{r}_v\|\,du\,dv$